

Scenario 2: Dynamic Residual Risk with Fixed Gamma

Detailed MSSTNB simulation validation report

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1 Overview

Scenario 2 evaluates whether the MSSTNB dynamic residual-risk component works when the temporal discount factor is treated as known. It is designed as a controlled extension of Scenario 1.

Scenario 1 used a clean baseline setting in which the residual risk was fixed at one. Scenario 2 changes only the residual-risk component: the true residual risk path is now dynamic, and

the fitted model samples this path. The temporal discount factor is fixed at the true value, $\gamma = 0.80$, so this scenario does **not** test discount-factor learning. Instead, it isolates the following question:

If the temporal smoothing level is known, can the model recover both the regression effects and the dynamic residual-risk path?

This separation is important because discount-factor learning and latent path recovery are different problems. If this scenario performs well, then later failures in a full model with unknown γ should not automatically be blamed on the FFBS path-recovery mechanism. They may instead reflect the weaker identifiability of γ .

The main conclusion is positive. Across 20 replicates, the model accurately recovers the regression effects and recovers the broad level pattern of the dynamic residual-risk path. The average log-lambda RMSE is 0.0468, the average 95% log-lambda coverage is 0.949, and the average correlation between posterior mean and truth for $\log \lambda$ is 0.787. Short-term temporal increments are harder to recover, with average correlation 0.188 for $\Delta \log \lambda$.

2 Scenario design

For region $j = 1, \dots, n_1$ and time $t = 1, \dots, T$, the coarsest-level observation model is summarized as

$$Y_{tj} \sim \text{NB}(\mu_{tj}, r_j),$$

where

$$\mu_{tj} = e_{tj} \lambda_{tj} \exp\{\beta_0 + \beta_1 x_{1,tj} + \beta_2 x_{2,tj} + \phi_j\}.$$

Here e_{tj} is the offset, $\beta_0, \beta_1, \beta_2$ are regression parameters, ϕ_j is the spatial random effect, r_j is the dispersion parameter, and λ_{tj} is the dynamic residual risk.

Scenario 2 differs from Scenario 1 only in the residual-risk component:

λ_{tj} evolves dynamically over time.

The temporal discount factor is fixed during fitting:

$$\gamma = \gamma_{\text{truth}} = 0.80.$$

Thus, the MCMC updates β , ϕ , τ_ϕ , r , κ , and λ , but it does not update γ , δ , or ω .

The conceptual logic is:

- Scenario 1 asks whether the regression, spatial, and dispersion components work in a baseline setting.
- Scenario 2 asks whether dynamic residual-risk recovery works when γ is fixed.
- A later full dynamic scenario can then ask whether γ itself can be learned.

3 Simulation and fitting settings

Table 1: Scenario 2 simulation and fitting settings.

Setting	Value
Scenario ID	S2_DYNAMIC_FIXED_GAMMA_T100
Number of replicates	20
Number of time points	100
Number of regions	9
Gamma in fitting model	0.80
x2 mode	continuous_time
MCMC iterations	40000
Burn in	20000
Thinning	5

The use of `continuous_time` for x_2 is deliberate. Earlier pilot checks showed that a region-level binary x_2 can align with the ICAR spatial effect in a small $n_1 = 9$ lattice, creating artificial spatial confounding. The formal Scenario 2 therefore uses a time-varying continuous x_2 , matching the clean comparison logic of Scenario 1 while adding only the dynamic residual-risk path.

4 Covariate and latent effect separation

Before interpreting regression recovery, we first check whether the simulated covariates are accidentally aligned with the latent components. This is a necessary diagnostic because the mean model contains the product

$$\exp(\beta_2 x_{2,tj}) \lambda_{tj},$$

and the additive log-risk term

$$\beta_2 x_{2,tj} + \phi_j + \log \lambda_{tj}.$$

If x_2 is correlated with ϕ_j or $\log \lambda_{tj}$, then regression recovery can look poor even if the sampler is correct. This is especially important when there are only nine regions.

Table 2: Confounding diagnostics across replicates.

Metric	Value
mean_cor_x1_x2	0.0018
min_cor_x1_x2	-0.0842
max_cor_x1_x2	0.0680
mean_cor_x2_loglambda	0.0007
min_cor_x2_loglambda	-0.0881
max_cor_x2_loglambda	0.1260
mean_cor_x2_phi	0.0021
min_cor_x2_phi	-0.1463
max_cor_x2_phi	0.1191

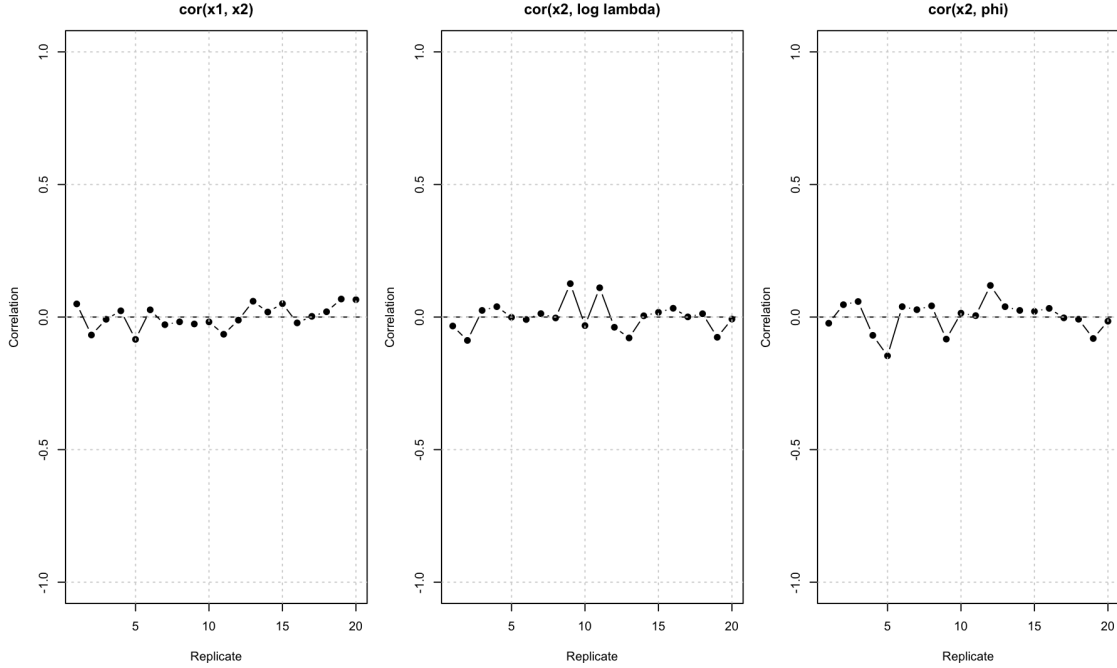


Figure 1: Scenario 2 confounding diagnostics across replicates.

The diagnostic results are clean. The average correlation between x_1 and x_2 is 0.0018, with

range $[-0.084, 0.068]$. The average correlation between x_2 and $\log \lambda$ is 0.0007, with range $[-0.088, 0.126]$. The average correlation between x_2 and ϕ is 0.0021, with range $[-0.146, 0.119]$.

These values are centered near zero. Therefore, the formal Scenario 2 datasets do not exhibit the artificial confounding that appeared in the earlier binary-region pilot. This gives a clean basis for evaluating whether the dynamic latent path itself harms regression recovery.

5 Regression parameter recovery

Regression recovery is evaluated by comparing posterior means and 95% credible intervals against the replicate-specific identified truth. The intercept truth varies across replicates because the latent residual-risk scale is identified through a recentering convention. This means β_0 should be compared to the identified truth reported in the summary table, not simply to the raw data-generating value before recentering.

Table 3: Regression parameter recovery across 20 replicates.

parameter	truth	treath	truth	mean	of post	mean	post	mean	post	mean	bias	cov-er	mean_post_sd
beta0	-	-	-	-1.5298	-1.5185	-	-	-	0.0083	1.00	0.0093		
	1.5295	1.7327	1.3944			1.7422	1.4007	0.0003					
beta1.beta1	0.5000	0.5000	0.5000	0.5034	0.5050	0.4878	0.5303	0.0034	0.0115	0.95	0.0102		
beta2.beta2-	-	-	-	-0.4015	-0.4005	-	-	-	0.0082	1.00	0.0098		
	0.4000	0.4000	0.4000			0.4156	0.3859	0.0015					

The regression recovery is strong. For β_1 , the posterior means average NA compared with truth NA. The mean bias is NA, the RMSE of the bias is NA, and coverage is NA.

For β_2 , the posterior means average NA compared with truth NA. The mean bias is NA, the RMSE of the bias is NA, and coverage is NA.

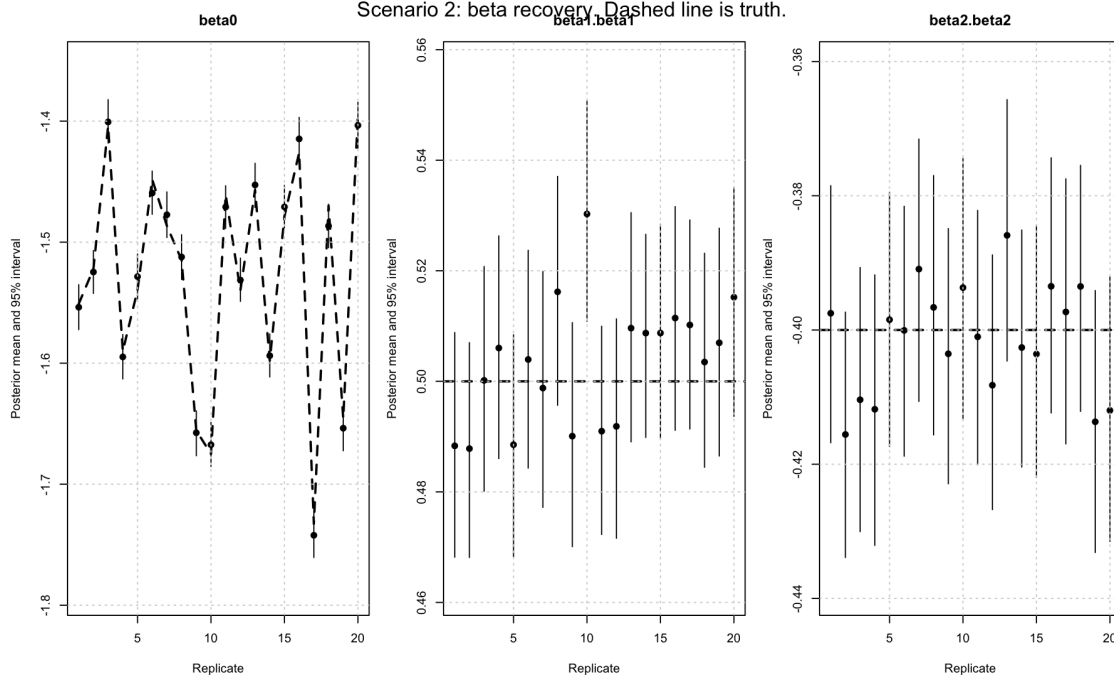


Figure 2: Posterior intervals for regression parameters across replicates.

This figure should be read as an interval recovery plot. Each interval is a posterior 95% credible interval for a replicate-specific parameter, and the reference point is the corresponding identified truth. The main question is whether the intervals are centered near the truth and whether they cover the truth at a reasonable frequency. The figure shows that the intervals for β_1 and β_2 are tightly centered near their true values. The intercept intervals are also centered near the identified intercept truth.

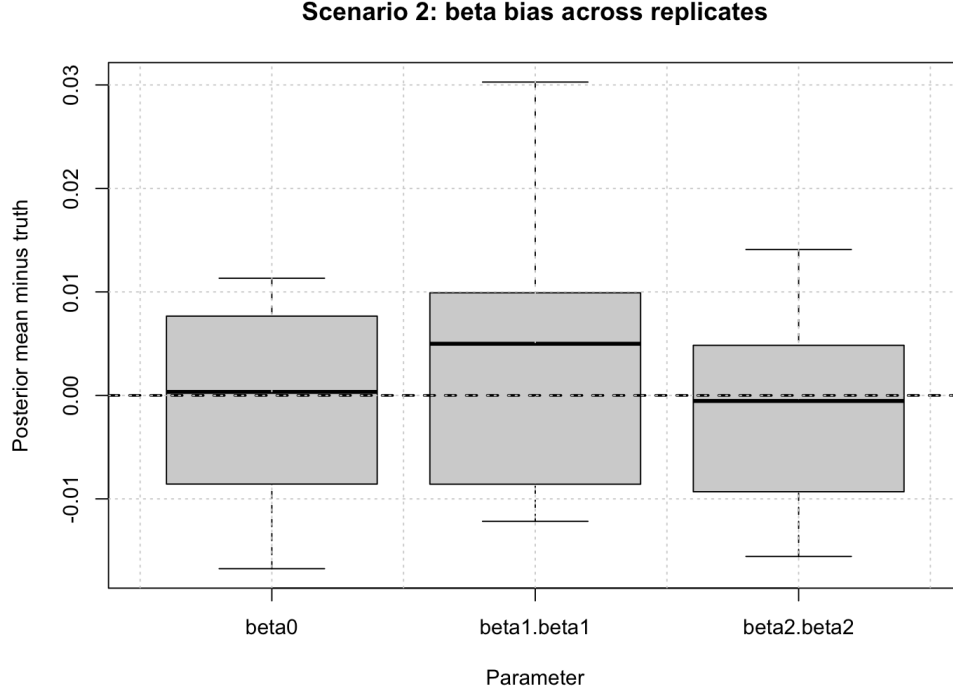


Figure 3: Bias distribution for regression parameters across replicates.

The bias boxplot gives the same conclusion in a more compact way. Biases are centered close to zero for all regression parameters. The average bias is -0.0003 for β_0 , NA for β_1 , and NA for β_2 .

The key interpretation is that adding dynamic residual risk does not compromise regression recovery in this controlled setting. This is a strong result because the fitted model now contains a flexible latent multiplicative path λ_{tj} that could, in principle, absorb covariate signal. The clean x_2 design and fixed γ make the regression effects identifiable enough to be recovered.

6 Dynamic residual-risk path recovery

The main target of Scenario 2 is the dynamic residual-risk path. Since λ_{tj} is positive and multiplicative in the mean model, path recovery is evaluated primarily on the log scale.

For each replicate, the main metric is

$$\text{RMSE}_{\log \lambda} = \left[\frac{1}{Tn_1} \sum_{t=1}^T \sum_{j=1}^{n_1} \{\log \hat{\lambda}_{tj} - \log \lambda_{tj}^{\text{truth}}\}^2 \right]^{1/2}.$$

We also report 95% credible interval coverage for $\log \lambda_{tj}$ and the correlation between posterior mean and truth:

$$\text{Cor}\{\log \hat{\lambda}_{tj}, \log \lambda_{tj}^{\text{truth}}\}.$$

Finally, we consider temporal increments,

$$\Delta \log \lambda_{tj} = \log \lambda_{tj} - \log \lambda_{t-1,j}.$$

This increment target is harder because it asks whether the model captures short-term increases and decreases rather than the broad level of the latent risk path.

Table 4: Dynamic residual-risk path recovery across 20 replicates.

Metric	Value
min_log_lambda_rmse	0.0342
mean_log_lambda_rmse	0.0468
median_log_lambda_rmse	0.0476
max_log_lambda_rmse	0.0573
mean_log_lambda_coverage_95	0.9494
median_log_lambda_coverage_95	0.9522
min_log_lambda_coverage_95	0.8967
max_log_lambda_coverage_95	0.9989
mean_cor_log_lambda	0.7866
median_cor_log_lambda	0.7880
min_cor_log_lambda	0.5382
max_cor_log_lambda	0.9133
mean_cor_delta_log_lambda	0.1880
median_cor_delta_log_lambda	0.1872
min_cor_delta_log_lambda	0.0749
max_cor_delta_log_lambda	0.2624

The level recovery is good. The mean log-lambda RMSE is 0.0468, with range [0.0342, 0.0573]. The mean 95% coverage for $\log \lambda_{tj}$ is 0.949, with range [0.897, 0.999]. The mean correlation between posterior mean and true $\log \lambda$ is 0.787, with range [0.538, 0.913].

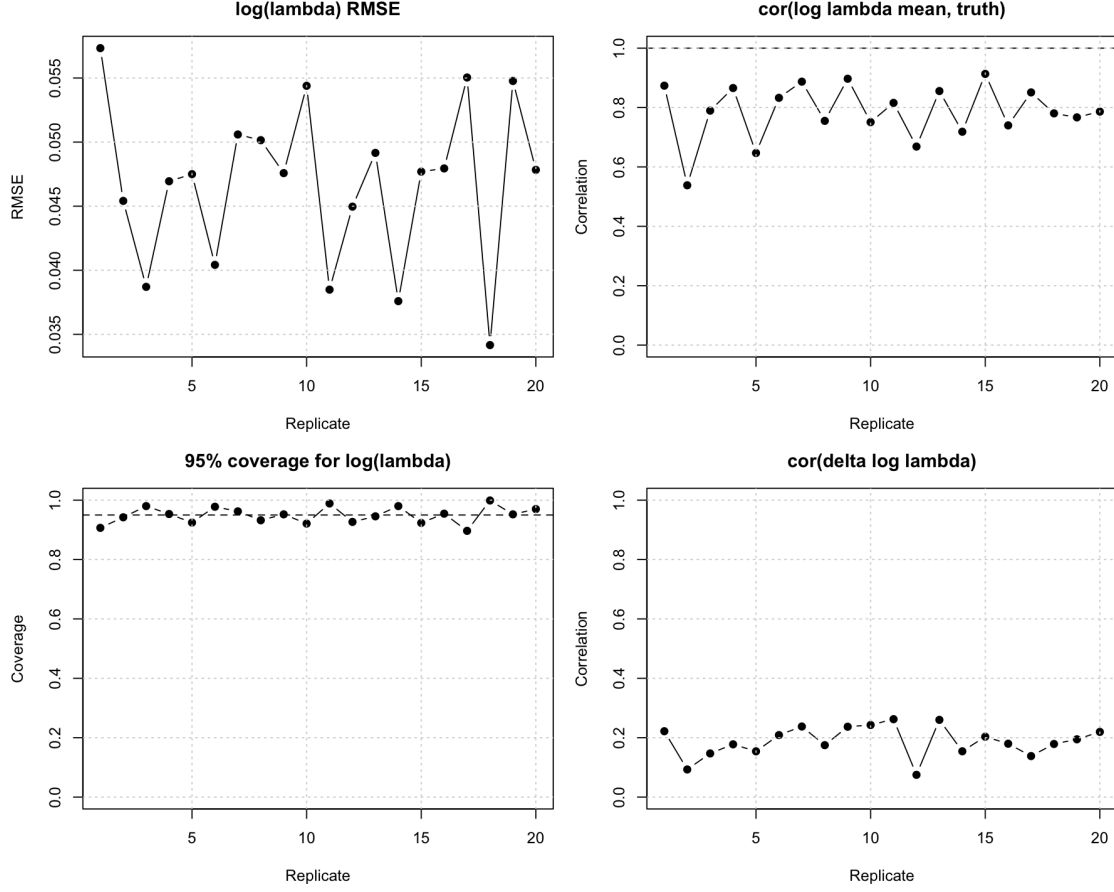


Figure 4: Dynamic residual-risk path recovery metrics across replicates.

This figure summarizes path recovery replicate by replicate. The RMSE panel shows that log-lambda errors are consistently small. The coverage panel shows that empirical coverage is close to the nominal 95% target. The correlation panel shows that the posterior mean path tracks the broad true path well in most replicates.

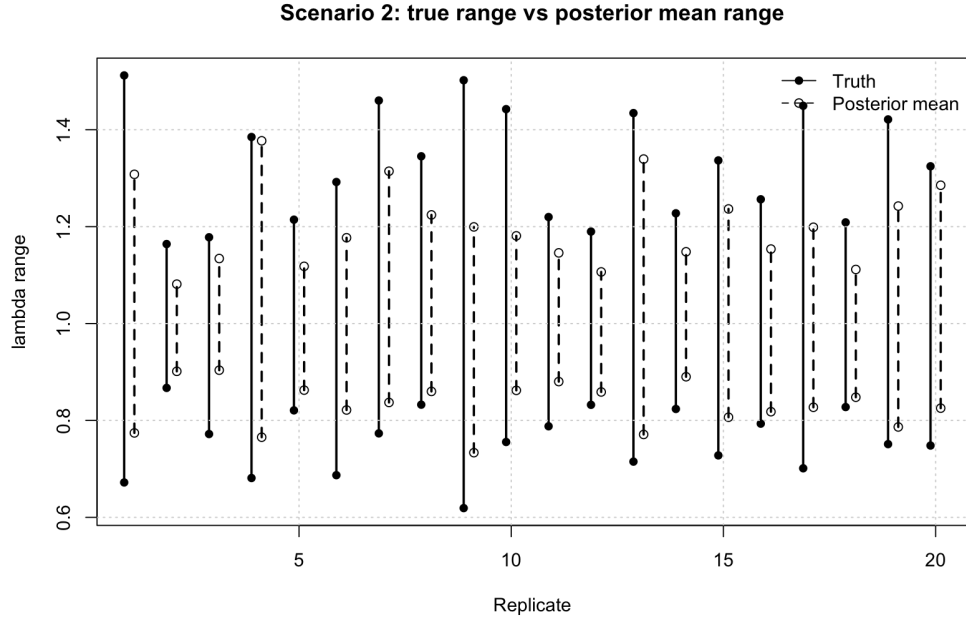


Figure 5: Range shrinkage from true lambda paths to posterior mean lambda paths.

The range shrinkage figure shows an important feature of latent path estimation. Posterior mean paths are smoother and less extreme than the true latent paths. This is expected: posterior means average over uncertainty, and the fixed-gamma dynamic evolution imposes temporal smoothing. Therefore, the model can recover the broad level of λ_{tj} while still shrinking extreme peaks and troughs toward the center.

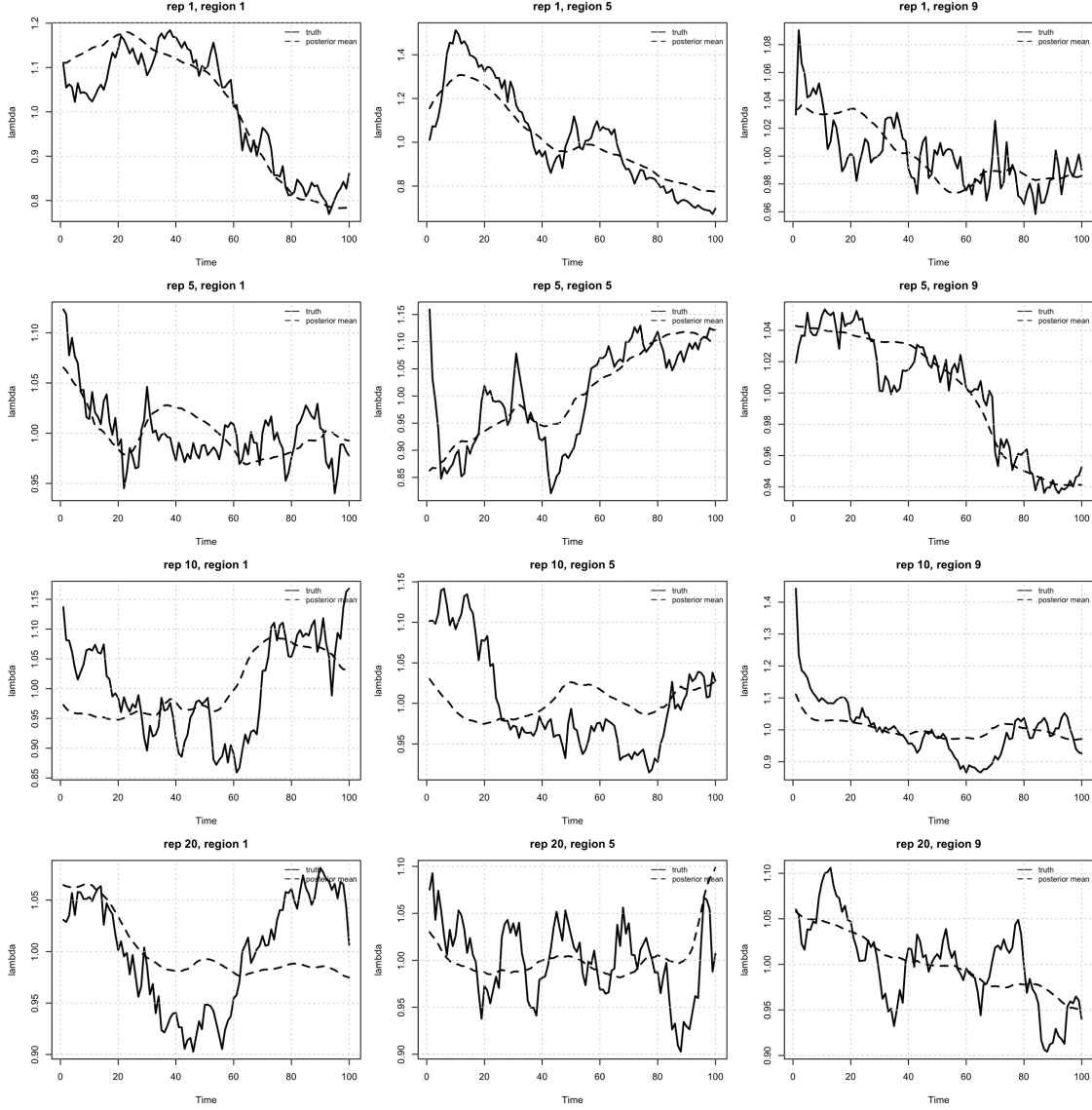


Figure 6: Representative dynamic residual-risk path recovery plots.

The representative path plots provide the most direct visual diagnostic. The true dynamic path and posterior mean path should move together over time, while the credible band should cover the true path reasonably often. The plots typically show good recovery of the broad path level and slower-moving trends. They also show that abrupt short-term wiggles are harder to recover.

This distinction is quantified by the temporal increment metric. The mean correlation for $\Delta \log \lambda$ is only 0.188, with range [0.075, 0.262]. This is much lower than the correlation

for $\log \lambda$ itself. Thus, Scenario 2 supports a nuanced conclusion: the model recovers the dynamic residual-risk level well, but it is less reliable for identifying exact one-step increases and decreases.

7 Spatial and dispersion recovery

Although Scenario 2 focuses on dynamic path recovery, the spatial and dispersion components should remain stable. If dynamic λ_{tj} disrupted the rest of the model, we would expect degraded recovery of ϕ_j or r_j .

Table 5: Spatial and dispersion recovery summaries.

Metric	Value
mean_phi_rmse	0.0251
median_phi_rmse	0.0234
mean_phi_cor	0.9968
median_phi_cor	0.9976
mean_phi_coverage	0.9556
mean_r_rmse	2.4103
median_r_rmse	2.5681
mean_r_bias	0.4693
mean_r_coverage	0.9611

The spatial summaries show the typical difficulty of estimating a small-area random effect with only 9 regions, but the results are stable enough for this validation setting. The mean spatial RMSE is 0.025, and the mean correlation between posterior mean and true ϕ is 0.997. These quantities should be interpreted as supportive diagnostics rather than the main target of Scenario 2.

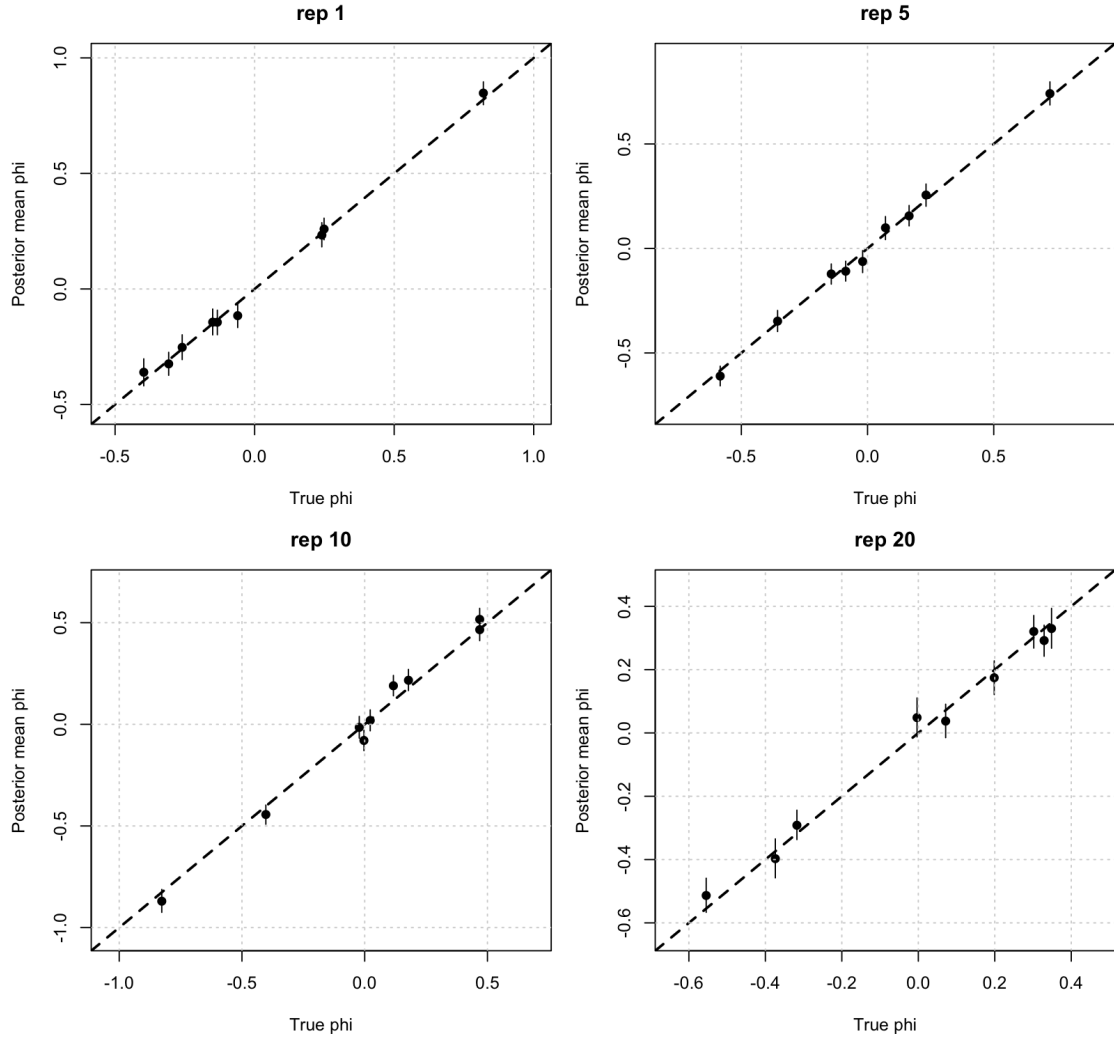


Figure 7: Spatial effect recovery summaries.

The truth-versus-posterior-mean plot checks whether estimated spatial effects align with the true spatial effects. Points near the diagonal indicate good spatial recovery. Some shrinkage is expected because the ICAR prior regularizes the spatial surface.

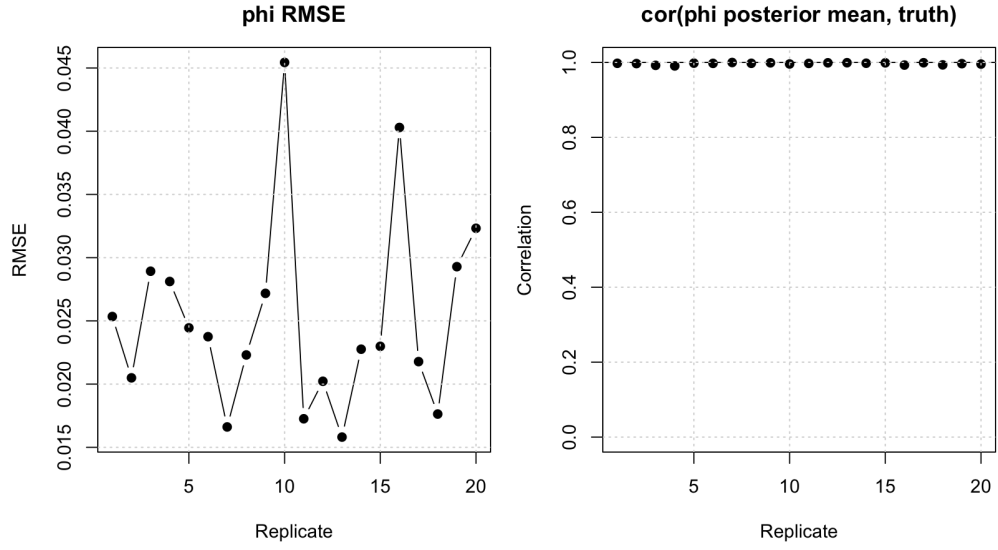


Figure 8: Spatial effect recovery metrics across replicates.

The spatial metric figure summarizes this behavior across replicates. The main purpose is to confirm that the dynamic residual-risk component does not cause spatial recovery to collapse.

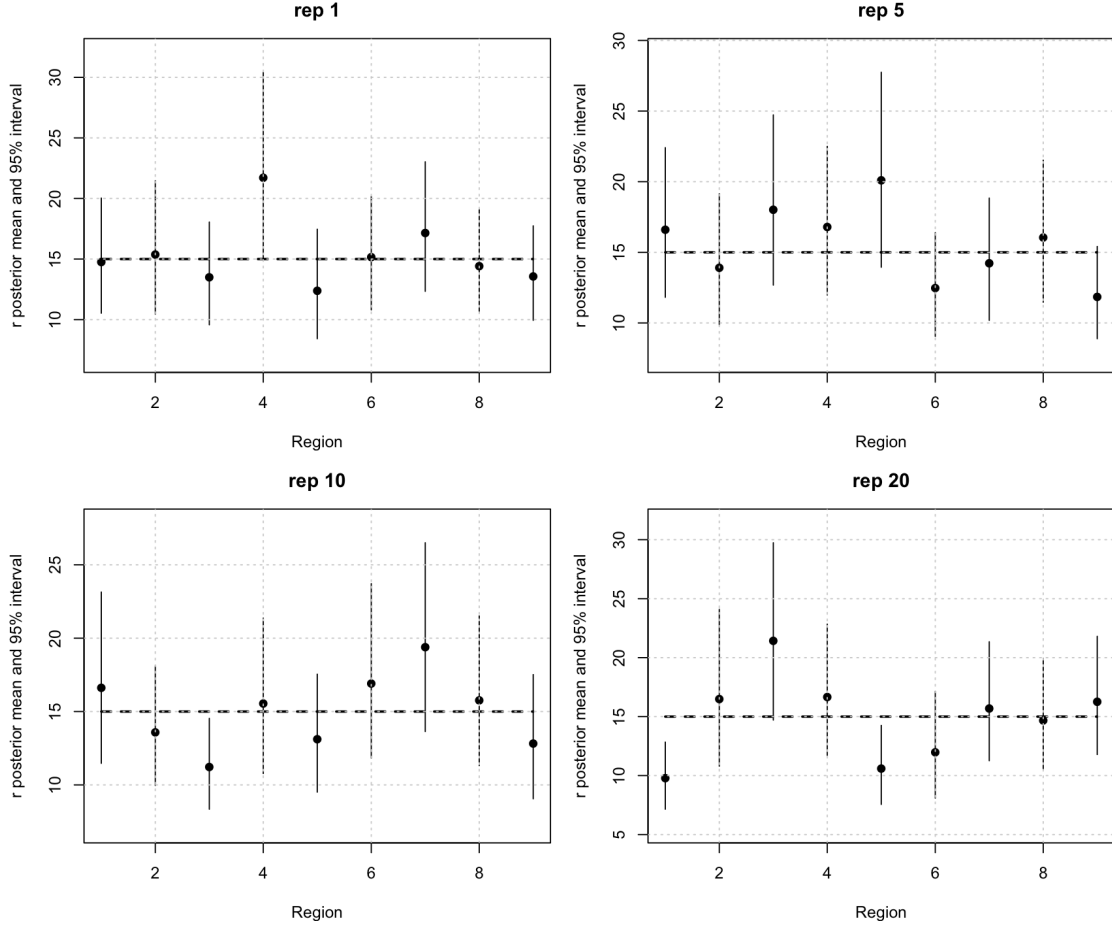


Figure 9: Dispersion parameter recovery across replicates.

The dispersion interval plot compares posterior intervals for r_j against the true dispersion values. Since r_j controls negative-binomial overdispersion, reasonable recovery indicates that the model is not forcing all extra-Poisson variation into either r_j or λ_{tj} alone.

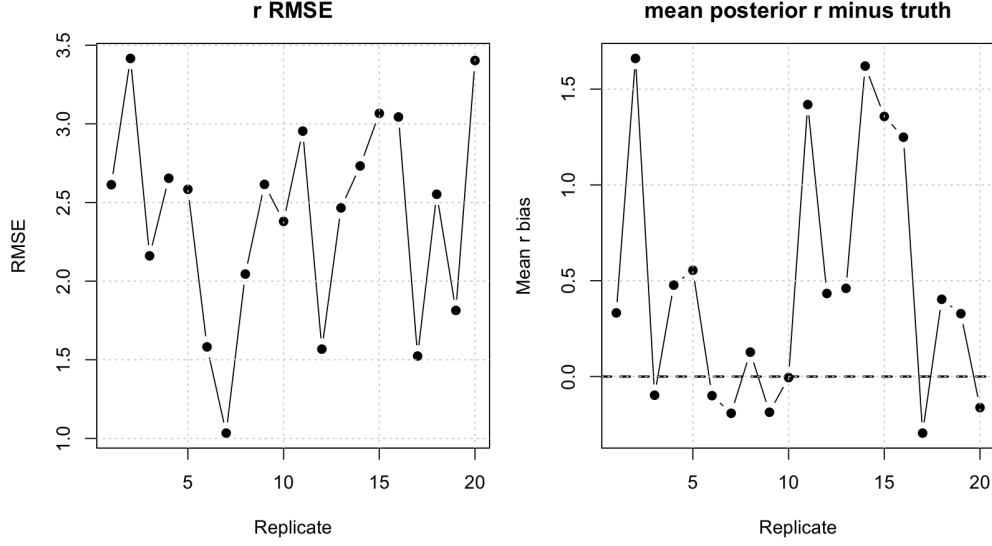


Figure 10: Dispersion recovery metrics across replicates.

The dispersion metrics confirm that r remains stable in this dynamic residual-risk setting. The mean r bias is 0.469, and the mean r coverage is 0.961.

8 MCMC diagnostics

The primary MCMC diagnostics are acceptance rates for the spatial and dispersion updates, rejection summaries for the beta elliptical slice step, and trace plots for selected parameters. These diagnostics check whether the formal batch results are computationally credible.

Table 6: MCMC diagnostic summaries across replicates.

Metric	Value
mean_phi_accept_rate	0.2548
median_phi_accept_rate	0.2520
mean_r_accept_rate	0.3030
median_r_accept_rate	0.3032
mean_beta_rejections	14.4984
median_beta_rejections	14.4848

The mean spatial acceptance rate is 0.255, and the mean dispersion acceptance rate is 0.303. These are in a reasonable range for random-walk Metropolis updates in this model. The beta elliptical slice step has mean rejection count 14.50, which is stable across the formal run.

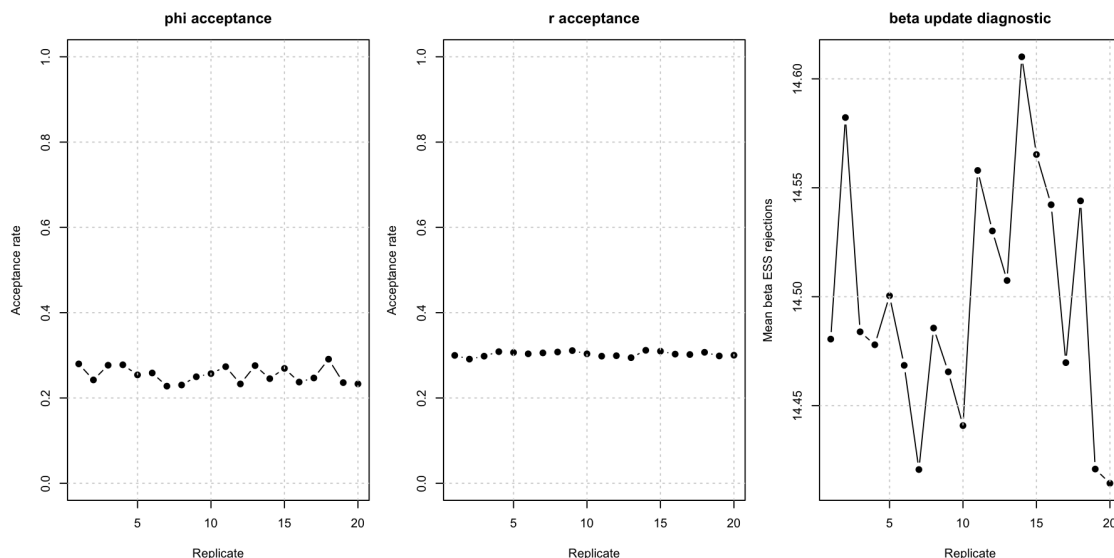


Figure 11: Main MCMC diagnostic summaries from Scenario 2.

The main MCMC diagnostic figure summarizes acceptance and rejection behavior across replicates. Its role is to catch obvious computational pathologies such as near-zero acceptance, excessive instability across replicates, or unusually poor beta elliptical slice behavior.

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The trace and ACF plots provide a more detailed check for selected replicates and selected parameters. These are not the primary scientific figures, but they are important for validating the simulation. The beta traces should fluctuate around stable levels without sustained drift. The log-likelihood traces should not show long-term trends after burn-in. The selected λ traces should show posterior movement without degenerating to a constant path.

9 Summary and interpretation

Scenario 2 successfully extends the Scenario 1 validation to a dynamic residual-risk setting. The results support four main conclusions.

First, regression recovery remains accurate when dynamic residual risk is introduced and γ is fixed at its true value. Across 20 replicates, the mean biases are close to zero for all regression parameters. In particular, β_2 has mean posterior mean NA, truth NA, RMSE bias NA, and coverage NA.

Second, dynamic residual-risk level recovery is strong. The posterior mean path tracks the true log-lambda path with mean correlation 0.787, and the mean log-lambda coverage is 0.949. This supports the central purpose of Scenario 2: when discount-factor learning is removed from the problem, the FFBS dynamic path mechanism can recover the broad latent residual-risk pattern.

Third, short-term temporal increments are harder to recover. The average correlation for $\Delta \log \lambda$ is 0.188, much lower than the level correlation. This should not be interpreted as a failure of the model. It means that the model is better suited to recovering smooth latent risk levels than exact one-step movements.

Fourth, the covariate design is clean. The correlations among x_2 , ϕ , and $\log \lambda$ are centered near zero, confirming that the formal Scenario 2 results are not driven by artificial covariate-latent confounding.

Overall, Scenario 2 validates the dynamic residual-risk path recovery mechanism under known temporal smoothing. Together with Scenario 1, it supports the staged simulation strategy: first validate baseline regression and spatial recovery, then validate dynamic latent path recovery under fixed γ , and only afterward evaluate the harder problem of learning γ from the data.